Let

$$\mathcal{A}_{n-1} = \{x_i = x_j\} \subset \mathbb{R}^n / \langle (1, \dots, 1) \rangle$$

be the type A_{n-1} braid arrangement. Its chambers are indexed by the $n!$ linear orders of the coordinates $x_1, \dots, x_n$.

There is a cyclic ordering of all chambers, obtained from Zaks' recursive Hamiltonian-cycle algorithm, with the following property. If

$$\Phi_n : \{\text{chambers of } \mathcal{A}_{n-1}\} \longrightarrow \mathbb{Z}/n!\mathbb{Z}$$

denotes this ordering, then the cyclic rotation of the indices

$$r : 1 \mapsto 2 \mapsto \cdots \mapsto n \mapsto 1$$

acts by

$$\Phi_n(r \cdot C) = \Phi_n(C) + (n-1)! \pmod{n!},$$

and the reversal of the indices acts by

$$\Phi_n(s \cdot C) = a - \Phi_n(C) \pmod{n!}$$

for some constant a.

Thus the natural dihedral subgroup $D_n \subset S_n$, acting on the indices $1, \dots, n$, is realized as the ordinary dihedral symmetry of the cycle of length $n!$ formed by all chambers.